5.5 For each of the following two time-series regression models, and assuming MR1–MR6 hold, find $\text{var}(b_2|\mathbf{x})$ and examine whether the least squares estimator is consistent by checking whether $\lim_{T\to\infty}\text{var}(b_2|\mathbf{x}) = 0$.

a. $y_t = \beta_1 + \beta_2 t + e_t, t = 1, 2, \ldots, T$. Note that $\mathbf{x} = (1, 2, \ldots, T)$, $\sum_{t=1}^{T}(t-\bar{t})^2 = \sum_{t=1}^{T} t^2 - \left(\sum_{t=1}^{T} t\right)^2 / T$, $\sum_{t=1}^{T} t = T(T+1)/2$ and $\sum_{t=1}^{T} t^2 = T(T+1)(2T+1)/6$.

b. $y_t = \beta_1 + \beta_2 (0.5)^t + e_t, t = 1, 2, \ldots, T$. Here, $\mathbf{x} = (0.5, 0.5^2, \ldots, 0.5^T)$. Note that the sum of a geometric progression with first term r and common ratio r is

$$S = r + r^2 + r^3 + \cdots + r^n = \frac{r(1-r^n)}{1-r}$$

c. Provide an intuitive explanation for these results.

a.

$$\text{var}(\text{b}_2|\mathbf{x}) = \frac{\sigma^2}{\sum_{t=1}^{T}(t-\bar{x})^2} = \frac{\sigma^2}{\sum_{t=1}^{T} t^2 - \frac{(\sum_{t=1}^{T} t)^2}{T}}$$

$$= \frac{\sigma^2}{\frac{T(T+1)(2T+1)}{6} - \frac{(T(T+1)/2)^2}{T}}$$

$$= \frac{\sigma^2}{\frac{2T^3+3T^2+T}{6} - \frac{T^3+T+2T^2}{4}}$$

$$= \frac{\sigma^2}{\frac{4T^3+6T^2+2T}{12} - \frac{3T^3+3T+6T^2}{12}} = \frac{12\sigma^2}{T^3-T}$$

$$\lim_{T\to\infty} var(b_2|\boldsymbol{x}) = \lim_{T\to\infty}\frac{12\sigma^2}{T^3-T} = 0(\text{consistent})$$

b.

$$\text{var}(\text{b}_2|\mathbf{x})$$

$$= \frac{\sigma^2}{\sum_{t=1}^{T}(0.5^t-\bar{x})^2} = \frac{\sigma^2}{\sum_{t=1}^{T} 0.5^{2t} - \frac{(\sum_{t=1}^{T} 0.5^t)^2}{T}}$$

where

$$\sum_{t=1}^{T} 0.5^{2t} = \sum_{t=1}^{T} 0.25^t = \frac{0.25(1-0.25^T)}{1-0.25} = \frac{(1-0.25^T)}{3}$$

$$\frac{(\sum_{t=1}^{T} 0.5^t)^2}{T} = \frac{\left(\frac{0.5(1-0.5^T)}{1-0.5}\right)^2}{T} = \frac{(1-0.5^T)^2}{T}$$

$$\Rightarrow \text{var}(\text{b}_2|\mathbf{x}) = \frac{\sigma^2}{\frac{(1-0.25^T)}{3} - \frac{(1-0.5^T)^2}{T}}$$

$$\lim_{T\to\infty} var(b_2|\boldsymbol{x}) = \lim_{T\to\infty} \frac{\sigma^2}{\frac{(1-0.25^T)}{3} - \frac{(1-0.5^T)^2}{T}} = \frac{\sigma^2}{\frac{1}{3} - \frac{1}{\infty}} = 3\sigma^2 (not\ consistent)$$

c.

簡單的模型(a)果然很好預測，b2 只要 T 很大其變異數等於零，相對的(b)是指數函數真的很複雜呢。

改正：var 的關鍵在於分母：$\sum(x_i - \bar{x})^2$，(a)小題$x_i = t$這個式子會持續發散。但是(b)小題$x_i = 0.5^t$會隨者 t 的上升收斂至 0，因此 var 不會等於零。

5.6 Suppose that, from a sample of 63 observations, the least squares estimates and the corresponding estimated covariance matrix are given by

$$\begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ -1 \end{bmatrix} \qquad \widehat{\text{cov}}(b_1, b_2, b_3) = \begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}$$

Using a 5% significance level, and an alternative hypothesis that the equality does not hold, test each of the following null hypotheses:

a. $\beta_2 = 0$
b. $\beta_1 + 2\beta_2 = 5$
c. $\beta_1 - \beta_2 + \beta_3 = 4$

a.

$$\text{H}_0: \beta_2 = 0\ vs. H_1: \beta_2 \neq 0$$
$$\text{t}_{97.5\%, 63-3} = 2$$
$$\text{rejection region: } \{\text{t}: |\text{t}| > 2\}$$
$$\text{t} = \frac{\text{b2} - \beta_2}{se(b2)} = \frac{3-0}{\sqrt{4}} = 1.5$$

Conclusion: we do not reject H_0

b.

$$\text{H}_0: \beta_1 + 2\beta_2 = 5\ \ vs.\ \ H_1: \beta_1 + 2\beta_2 \neq 5$$
$$\text{t}_{97.5\%, 63-3} = 2$$
$$\text{rejection region: } \{\text{t}: |\text{t}| > 2\}$$
$$\widehat{\text{var}}(\text{b}_1 + 2\text{b}_2) = [1,2] \begin{bmatrix} 3 & -2 \\ -2 & 4 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 11$$
$$\text{t} = \frac{\text{b}_1 + 2\text{b}_2 - 5}{se(\text{b}_1 + 2\text{b}_2)} = \frac{2 + 2*3 - 5}{\sqrt{11}} = 0.905$$

Conclusion: we do not reject H_0

c.

$$\text{H}_0: \beta_1 - \beta_2 + \beta_3 = 4\ \ vs.\ \ \beta_1 - \beta_2 + \beta_3 \neq 4$$
$$\text{t}_{97.5\%, 63-3} = 2$$
$$\text{rejection region: } \{\text{t}: |\text{t}| > 2\}$$

$$\widehat{\text{var}}(b_1 - b_2 + b_3) = [1,-1,1]\begin{bmatrix} 3 & -2 & 1 \\ -2 & 4 & 0 \\ 1 & 0 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = [6,-6,4]\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 16$$

$$t = \frac{b_1 - b_2 + b_3 - 4}{se(b_1 - b_2 + b_3)} = \frac{2-3-1-4}{\sqrt{16}} = -1.5$$

Conclusion: we do not reject H_0

5.20 A generalized version of the estimator for β_2 proposed in Exercise 2.9 by Professor I.M. Mean for the regression model $y_i = \beta_1 + \beta_2 x_i + e_i, i = 1, 2, \ldots, N$ is

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

where $(\bar{y}_1, \bar{x}_1)$ and $(\bar{y}_2, \bar{x}_2)$ are the sample means for the first and second halves of the sample observations, respectively, after ordering the observations according to increasing values of x. Given that assumptions MR1–MR6 hold:

a. Show that $\hat{\beta}_{2,mean}$ is unbiased.

b. Derive an expression for $\text{var}\left(\hat{\beta}_{2,mean}|\mathbf{x}\right)$.

c. Write down an expression for $\text{var}\left(\hat{\beta}_{2,mean}\right)$.

d. Under what conditions will $\hat{\beta}_{2,mean}$ be a consistent estimator for β_2?

e. Randomly generate observations on x from a uniform distribution on the interval (0,10) for sample sizes $N = 100, 500, 1000$, and, if your software permits, $N = 5000$. Assuming $\sigma^2 = 1000$, for each sample size, compute:

 i. $\text{var}(b_2|\mathbf{x})$ and $\text{var}\left(\hat{\beta}_{2,mean}|\mathbf{x}\right)$ where b_2 is the OLS estimator.

 ii. Estimates for $E\left[\left(s_x^2\right)^{-1}\right]$ and $E\left[4/\left(\bar{x}_2 - \bar{x}_1\right)^2\right]$ where s_x^2 is the sample standard deviation for x using N as a divisor.

f. Comment on the relative magnitudes of your answers in part (e), (i) and (ii) and how they change as sample size increases. Does it appear that $\hat{\beta}_{2,mean}$ is consistent?

g. Show that $E\left[\left(s_x^2\right)^{-1}\right] \xrightarrow{p} 0.12$ and $E\left[4/\left(\bar{x}_2 - \bar{x}_1\right)^2\right] \xrightarrow{p} 0.16$. [*Hint*: The variance of a uniform random variable defined on the interval (a, b) is $(b - a)^2/12$.]

h. Suppose that the observations on x were not ordered according to increasing magnitude but were randomly assigned to any position. Would the estimator $\hat{\beta}_{2,mean}$ be consistent? Why or why not?

a.

$$\hat{\beta}_{2,mean} = \frac{\bar{y}_2 - \bar{y}_1}{\bar{x}_2 - \bar{x}_1}$$

$$\mathrm{E}\left(\hat{\beta}_{2,mean}\right)=\mathrm{E}\left(\frac{\bar{y}_2-\bar{y}_1}{\bar{x}_2-\bar{x}_1}\right)=\mathrm{E}\left(\frac{\frac{2}{N}\left(\sum_{\frac{N}{2}+1}^{N}y_i-\sum_{1}^{\frac{N}{2}}y_i\right)}{\frac{2}{N}\left(\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i\right)}\right)$$

$$=\mathrm{E}\left(\frac{\sum_{\frac{N}{2}+1}^{N}(\beta_1+\beta_2x_i+e_i)-\sum_{1}^{\frac{N}{2}}(\beta_1+\beta_2x_i+e_i)}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}\right)$$

$$=E\left(\frac{\beta_2\left(\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i\right)+\sum_{\frac{N}{2}+1}^{N}e_i-\sum_{1}^{\frac{N}{2}}e_i}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}\right)$$

$$=E\left(\frac{\beta_2\left(\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i\right)}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}\right)+E\left(\frac{\sum_{\frac{N}{2}+1}^{N}e_i}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}\right)$$

$$-E\left(\frac{\sum_{1}^{\frac{N}{2}}e_i}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}\right)=\mathrm{E}(\beta_2)+0-0=\beta_2$$

b. **x** 和β_2都是常數

$$\mathrm{var}(\hat{\beta}_{2,mean}|\boldsymbol{x})=var\left(\frac{\beta_2\left(\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i\right)+\sum_{\frac{N}{2}+1}^{N}e_i-\sum_{1}^{\frac{N}{2}}e_i}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}|\boldsymbol{x}\right)=$$

$$\left(\frac{1}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}\right)^2var\left(\beta_2\left(\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i\right)+\sum_{\frac{N}{2}+1}^{N}e_i-\sum_{1}^{\frac{N}{2}}e_i\,|\boldsymbol{x}\right)=$$

$$\left(\frac{1}{\sum_{\frac{N}{2}+1}^{N}x_i-\sum_{1}^{\frac{N}{2}}x_i}\right)^2\left(var\left(\sum_{\frac{N}{2}+1}^{N}e_i\right)+var\left(\sum_{1}^{\frac{N}{2}}e_i\right)\right)=\left(\frac{\frac{2}{\mathrm{N}}}{\bar{x}_2-\bar{x}_1}\right)^2N\sigma^2=\frac{4\sigma^2}{N}\frac{1}{(\bar{x}_2-\bar{x}_1)^2}$$

c.

$$\mathrm{var}(\hat{\beta}_{2,\mathrm{mean}})=\mathrm{E}[\mathrm{var}(\hat{\beta}_{2,\mathrm{mean}}|\mathbf{x})]+\mathrm{var}(\mathrm{E}[\hat{\beta}_{2,\mathrm{mean}}|\mathbf{x}])$$

$$=\mathrm{E}(\left(\frac{\frac{2}{\mathrm{N}}}{\bar{x}_2-\bar{x}_1}\right)^2N\sigma^2)+\mathrm{var}(\beta_2)=\frac{4\sigma^2}{N}E\left(\frac{1}{(\bar{x}_2-\bar{x}_1)^2}\right)$$

d.

$$\lim_{N\to\infty} \mathrm{var}(\hat{\beta}_{2,\mathrm{mean}}) = \lim_{N\to\infty} \frac{4\sigma^2}{N} E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right) \text{一致性必須收斂到 } 0$$

因此一致性的條件就是$E\left(\frac{1}{(\bar{x}_2-\bar{x}_1)^2}\right)$必須收斂到一個常數

e.

(i)

100 var(b2)= 1.227853 var(B2)= 1.599313

500 var(b2)= 0.2512013 var(B2)= 0.3400983

1000 var(b2)= 0.1200279 var(B2)= 0.1570505

5000 var(b2)= 0.02380369 var(B2)= 0.03165616

(ii)

100 = 0.121834 = 0.164483

500 = 0.1201063 = 0.160391

1000 = 0.1201993 = 0.1604331

5000 = 0.1199233 = 0.1599155

f.

(i)小題，N 上升 var 會下降，但是(ii)小題幾乎不變。由於(i)小題的 var(B2)隨著樣本數增加而下降，可以說是 consistent

g.

var inverse of x is $\left(\frac{10^2}{12}\right)^{-1} = 0.12$

取期望值仍為 0.12

$$\bar{x}_2 = \frac{10+5}{2} = 7.5$$

$$\bar{x}_1 = \frac{5+0}{2} = 2.5$$

$$E\left(\frac{4}{(\bar{x}_2 - \bar{x}_1)^2}\right) = E\left(\frac{4}{(7.5-2.5)^2}\right) = \frac{4}{5^2} = 0.16$$

h.

$$\mathrm{var}(\hat{\beta}_{2,\mathrm{mean}}) = \frac{4\sigma^2}{N} E\left(\frac{1}{(\bar{x}_2 - \bar{x}_1)^2}\right)$$

沒有一致性。

不排序的話$(\bar{x}_2 - \bar{x}_1)$會等於 0，因此$E\left(\frac{1}{(\bar{x}_2-\bar{x}_1)^2}\right)$不是一個有限的數字，就無法讓$\mathrm{var}(\hat{\beta}_{2,\mathrm{mean}})$收斂到 0。